SIMATS ENGINEERING CO1 - ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA0403

Name of the Student	SHAM S.
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COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%
Assessment Tool Weightage: 35%
Duration: 1 Hour
Marks: 50

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. Coin Toss MLE

Given

Outcomes:
H H T H H H T H H T

Total tosses (**n**) = 10

Number of Heads (**k**) = 7

Number of Tails = 3

MLE Formula

For a Bernoulli/Binomial distribution,

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer

Estimated probability of Heads = 0.7 (70%)

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2. Students Passing Exam

Given

- Total students = 50
- Passed = 42

MLE Formula

$$\hat{p} = \frac{\text{Passed}}{\text{Total Students}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer

MLE of passing probability = 0.84 (84%)

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3. Defective Light Bulbs

Given

- Total bulbs = 100
- Defective = 8

MLE Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer

MLE of defective probability = 0.08 (8%)

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4. Biased Die

Given

- Total rolls = 60
- Number of sixes = 18

MLE Formula

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.3$$

Answer

Estimated probability of rolling a 6 = 0.3 (30%)

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5. Prediction, Error, and Weight Update

Given

- Input, $x = 4$
- Weight, $w = 2$
- Actual Output = 10

- Prediction formula:

$$y = w \times x$$

Learning rate:

$$\alpha = 0.01$$

Gradient:

$$\nabla = 4$$

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(1) Predicted Output

$$y = 2 \times 4 = 8$$

Answer

Predicted Output = 8

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(2) Calculate the Error

Error is calculated as:

$$\begin{aligned} &\text{Error} = \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2 \end{aligned}$$

Answer

Error = 2

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(3) Update the Weight

Gradient Descent update rule:

$$w_{\text{new}} = w - \alpha \times \text{Gradient}$$

Substitute the values:

$$\begin{aligned} & w_{\text{new}} = 2 - (0.01 \times 4) \\ & = 2 - 0.04 \\ & = 1.96 \end{aligned}$$

Answer

Updated Weight = 1.96

Code of Conduct:

I SHAM S. (192325076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Name of the student:*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation